PEARSON EDEXCEL IAL PSYCHOLOGY · PAPER 3 (WPS03/01) · TOPIC C

Chapter 19 — Practical Investigation

Diagnostic Training Workbook · 诊断训练册

Correlational study · Spearman's rank · V/R/G evaluation · Hikaru as running case

Name 姓名: ____________________ Class 班级: ______________ Start date: ____________

How to use this workbook 使用方法

1. **Workflow, not reference**. Each page mirrors one Step of the courseware. Do the step first, then fill in the page. 不是参考册,是配合课件 7-step workflow 的训练册。
2. **Hikaru is the running case**. Three pages (4, 6, 11) carry a diagnostic task — read Hikaru's scenario, then write your own criticism. Hikaru 是一个真实考试 scenario,每节诊断一次。
3. **Write out AMSPRC for your own practical** (Aim · Method · Sample · Procedure · Results · Conclusion) before looking at p.13 answers.
4. **Pomodoro**: 25 min on, 5 min off. Tick each pomodoro as you finish a page.

Pomodoro tracker

🍅 P1 🍅 P2 🍅 P3 🍅 P4 🍅 P5 🍅 P6 🍅 P7 🍅 P8 🍅 P9 🍅 P10 🍅 P11 🍅 P12 🍅 P13

Page tracker 完成度

1	2	3	4	5	6	7	8	9	10	11	12	13

What this chapter covers 大纲考点 (Spec 3.4)

Syllabus	Content	Workbook pages
Spec 3.4 SPEC	6 musts: correlational study + descriptive stats + non-parametric test; design decisions; scatter diagram; Spearman's + significance; V/R/G evaluation; report writing	p. 2
Step 1 STEP	Aim · hypothesis · operationalise co-variables	pp. 3–4
Step 2 STEP	Sampling · ethics (consent form) · controls · pilot	pp. 5–6
Step 3 STEP	Materials · 6-step procedure	p. 7
Step 4 STEP	Tabulate data · descriptive stats · scatter · Spearman's · significance	pp. 8–9
Step 5 V R G	Validity · Reliability · Generalisability	p. 10
Step 6 IMP	Improvements (each linked to a weakness)	p. 11
Step 7 SPEC	Procedure · Results · Discussion + 3 textbook exam Qs	p. 12
Brain dump	Free recall + answer key	p. 13

Courseware link 配套课件

Open in browser: `https://lawrenceliu007.github.io/edexcel-psychology-courseware/edexcel/ch19-practical/index.html`

Spec 3.4 musts · 7-step workflow

COURSES §1

Spec 3.4 — fill in the 6 musts

Read the courseware §1, then fill the 6 boxes in your own words.

1. Design + conduct a ____________ study on aggression or body rhythms, with ____________ statistics + a non-parametric test.
2. Make design decisions: ____________, operationalise, _________, formulate _________, controls.
3. Present findings with a ____________ diagram.
4. Conduct a _________ test and decide on statistical _________.
5. Evaluate _________ · _________ · _________.
6. Write up _________ · _________ · _________ of a report.

The 7-step workflow at a glance

Step	What you do	Workbook
0	Pick topic + understand exam format (Hikaru-style scenarios)	p. 2
1	Aim · hypothesis · operationalise co-variables	pp. 3–4
2	Sampling · consent form · controls · pilot study	pp. 5–6
3	Materials · 6-step procedure	p. 7
4	Tabulate data · descriptive stats · scatter · Spearman's · significance	pp. 8–9
5	V / R / G evaluation	p. 10
6	Improvements (linked to weaknesses)	p. 11
7	Report writing + 3 textbook exam questions	p. 12

The 5 syllabus learning objectives

1. Design and conduct a correlational study on ____________ or ____________.
2. Make design decisions when _________ and gathering data.
3. Analyse correlational data including an _________ test for a relationship.
4. Evaluate _________ and weaknesses of the correlation and suggest _________.
5. Write up the _________, _________ and _________ section of a report.

🎯 **Exam reality check**

- **No "design from scratch"** question. Every practical Q gives a scenario (Hikaru, Cyr, etc.). You patch the design or evaluate it. The exam tests **diagnostic ability**.
- **No need to bring notes**. Exam is closed-book. But you must remember your own design as a reference point when diagnosing scenarios.
- Treat your practical like a **named study**: examiner expects AMSPRC + V/R/G.

Pick your topic — circle one to commit to

AGGRESSION TRACK

- Masculinity × aggression (textbook n=10)
- Height × self-reported aggression
- Self-reported aggression × contact-sport injuries
- Premenstrual tension × self-reported aggression
- Age × self-reported aggression

BODY RHYTHMS TRACK

- Age × sleep duration
- Physical exercise × sleep quality
- Severity of premenstrual tension × hours of sleep
- Sleep quality × unhealthy snacking
- Social media time × sleep duration

My chosen topic: ____

Step 1 — Aim, hypothesis, variables

COURSES §2

Aim · hypotheses · operationalisation

Read §2 first. Use the fixed phrase "*To investigate whether there is a link between X and Y*" for the aim.

1 · My aim （一句话,To investigate whether...）

2 · My alternative hypothesis （方向性/directional or 非方向性/non-directional）

Directional Non-directional · （方向性配 one-tailed;非方向性配 two-tailed）

3 · My null hypothesis （永远写,不管方向性）

4 · Operationalising co-variable 1

Abstract concept	______________________
Measure used	______________________
Score range	______________________
Data level	（Must be at least ordinal） __________

5 · Operationalising co-variable 2

Abstract concept	______________________
Measure used	______________________
Score range	______________________
Data level	（Must be at least ordinal） __________

Verify against textbook pp.216-217

Masculinity × *aggression*: use a **computerised quiz** (0–100) + a **video game** (0–20, 1 pt per character). Cross-check your 4–5 answers against these.

💡 **Avoid these slips**

- Don't call them **IV / DV —** they are **co-variables**. Edexcel mark scheme downmarks IV/DV for a correlation.
- Directional → **one-tailed** Spearman's. Spearman's needs **ordinal** data minimum.

🔍 Diagnose Hikaru · Step 1

COURSES §2

Hikaru's scenario (running case study)

Scenario — Hikaru's practical

*Hikaru is planning a study to see whether eating sugary food affects aggressive behaviour in **12-year-old children**. She plans to give 100 children a sugary snack and then ask them **"Were you aggressive today?"** She will compare their yes/no answers to a teacher rating of aggression (1–5).*

Q1 — Data level mismatch

What kind of variable is "Were you aggressive today?" (Yes/no = ?)

What level of data does Spearman's rank need? (at least ?)

Will Spearman's rank work on Hikaru's yes/no answers? Why or why not?

Q2 — Hypothesis direction

Should Hikaru's hypothesis be directional or non-directional?

Defend your choice in one sentence (cite prior literature or absence)

Q3 — Ethics

Is 12 a suitable age for this study?

Which BPS ethical guideline is most at risk? (e.g. consent / protection / withdrawal / deception)

Q4 — Operationalisation (the deepest issue)

What does "aggressive behaviour" actually mean in Hikaru's study? Is it the same thing when measured by a child's yes/no vs a teacher's 1–5 rating?

🎯 **The point of this diagnostic**

Hikaru is a real exam-style scenario (WPS02/01 Q2). The exam **never** asks you to design from scratch — it gives a flawed scenario and asks you to spot the flaws. Practice spotting them on Hikaru now so you spot them in the exam room.

Step 2 — Design decisions

COURSES §3

Sampling · ethics · controls · pilot

Read §3 first. Fill in for your own topic.

1 · My sampling method (circle one)

Opportunity Volunteer Random Stratified

Why this method? (1 sentence defence)

2 · My sample

n (target)	____________
Target population	____________________________________
Recruitment route	(where you'll find them) ____________________________________

Consent form — 7 mandatory items (textbook p.218)

Tick the items your consent form will include.

- ☐ 1. **Aim(s)** of the study — what you are investigating
- ☐ 2. **Exactly what they will do** — full detail so they can decide
- ☐ 3. **Any possible implications** — e.g. playing a moderately aggressive game
- ☐ 4. **Right to withdraw** at any point without consequence + take data with them
- ☐ 5. **Anonymisation & destruction** — names → numbers, data destroyed after set date
- ☐ 6. **Who else sees findings** + how information will be distributed
- ☐ 7. **Contact details** (researcher + supervisor) + signatures

Controls (textbook p.217)

List the controls you will put in place. Aim for at least 4.

1. ____
2. ____
3. ____
4. ____

Special rule for aggression studies: **no one under 16**; game must be age-appropriate; participants told in advance about the violent content.

Is this rule relevant to my study? Yes · No — (explain)

Pilot study

Purpose of my pilot (in 1 sentence)

Pilot participants (how many, who)

💡 **Writing sampling weakness for AO3**

Don't just say "*small sample*". Link: **"opportunity sample → only represents students at this school → findings cannot be generalised to the wider male population."** The linkage is where AO3 marks come from.

🔍 Diagnose Hikaru · Step 2

COURSES §3

Scenario — Hikaru's recruitment and method

*Hikaru recruits 100 children from **one primary school** whose parents returned a permission slip. She gives them a sugary snack, then asks **"Were you aggressive today?"** (yes/no), then a teacher rates their aggression 1–5.*

Q1 — Sampling weakness

Name ONE sampling weakness in Hikaru's design.

How does this limit her findings? (link to generalisability)

Q2 — Ethical issue

Name ONE ethical issue Hikaru should have addressed before the children took part.

Which BPS principle does it violate? (e.g. informed consent / right to withdraw / protection from harm / debrief)

Q3 — Reliability of "Were you aggressive today?"

Why is asking a 12-year-old "Were you aggressive today?" an unreliable measure? (think about how a child would interpret "aggressive")

Q4 — Demand characteristics

After eating the sugary snack, do you think children might answer the question differently? Why?

Q5 — Improve Hikaru's design (preview of Step 6)

Suggest ONE improvement that would address the closed-question weakness. Be specific.

📝 **Self-check before moving on**

Did you write at least **2 lines per answer** and link to AO3 vocabulary (generalisability, validity, BPS principle, demand characteristics)? If yes, move to p.7.

Step 3 — Materials & procedure

COURSES §4

Measurement tools

Both co-variables must be at least ordinal. Choose one tool per variable.

Co-variable 1 tool

Questionnaire / rating scale Computerised quiz Behavioural measure

strength:

weakness:

Co-variable 2 tool

Questionnaire / rating scale Computerised quiz Behavioural measure

strength:

weakness:

6-step procedure (textbook p.219)

Write your own 6 steps. Each step should be 1-2 sentences with timings where relevant.

Step 1 · Briefing + informed consent (sign forms, allocate participant number)

Step 2 · First measurement (where, what tool, how long)

Step 3 · Waiting period (what happens between the two measures)

Step 4 · Second measurement (where, what tool, practice + data run)

Step 5 · Debrief (right to withdraw, questions, see results when available)

Step 6 · Thank + release (thanks, optional insights, dismiss)

Standardised instructions (write the opening script)

🎯 **This page feeds the 4-mark "Describe the procedure" Q**

- Your 6 steps must mention all 4: **Sample** (who, n, from where, technique) · **Apparatus & scoring** · **Design** (repeated measures) · **Procedure** (sequence + timings).
- Don't call it "matched pairs" or "matched samples" — it's **repeated measures** (both vars recorded for every participant).

Step 4 — Spearman's rank worked example

COURSES §5

Rank the textbook data (n=10)

Read §5 first. We will redo Table 19.3 from memory. Fill the ranks for both co-variables, then d².

Participant	1	2	3	4	5	6	7	8	9	10
Masculinity (A)	56	69	30	88	80	87	95	67	77	45
Aggression (B)	17	18	10	10	15	16	17	15	14	6
Rank A	____	____	____	____	____	____	____	____	____	____
Rank B	____	____	____	____	____	____	____	____	____	____
d²	____	____	____	____	____	____	____	____	____	____

Tied ranks — practice with aggression scores

Aggression scores in ascending order: 6, 10, 10, 14, 15, 15, 16, 17, 17, 18. Two 10s and two 15s and two 17s. Average the number positions they occupy.

Number position	1	2	3	4	5	6	7	8	9	10
Score B (ascending)	6	10	10	14	15	15	16	17	17	18
Rank for B	1	____	____	____	____	____	____	____	____	____

Calculate r_s

Plug into the formula. $r_s = 1 - (6 \times \Sigma d^2) \div (n(n^2 - 1))$

Σd² (sum of squared differences) = ________

n = ______

Calculation:

---

r_s = ____________ (expected: 0.318)

> 💡 **Tied ranks rule**
>
> If two scores are equal, average the number positions they occupy. Two 10s at positions 2 and 3 → both get (2+3)/2 = **2.5**. The next score (14) then gets rank **4** (skip 3).

Step 4 — Significance decision

COURSES §5

Significance decision (textbook p.222)

n	One-tailed p<0.05	One-tailed p<0.01	Two-tailed p<0.05	Two-tailed p<0.01
7	0.714	0.893	0.786	0.929
8	0.643	0.833	0.738	0.881
9	0.600	0.783	0.700	0.833
10	**0.564**	0.745	0.648	0.794

Worked decision (textbook)

calculated r_s = **0.318**; n = 10; one-tailed test at $p < 0.05$ → critical value = **0.564**.

0.318 < **0.564** → calculated is **less than** critical → **not significant** → **accept null hypothesis**.

Fill in for your own study

calculated r_s = ________; n = ____; one/two-tailed = ________; p < ___ → critical value = ________

Decision: **significant** or **not significant** · (circle one)

The 7 mandatory ingredients of a statistical conclusion

EXAM TIP p.223. A statistical statement must include all 7 to score full marks.

1. The ________________ value (e.g. "r_s = 0.318")
2. The __________ value (e.g. "0.564")
3. The probability / level of __________ (e.g. "$p < 0.05$")
4. The __________ of participants (e.g. "n = 10")
5. Whether it was a __________ or two-tailed test
6. Whether calculated was __________ or equal to critical
7. What __________ is supported (restate it)

Write the textbook's model conclusion (from memory first, then check p.13)

"The calculated value of the Spearman's rank test was r = ___..." — write the full sentence.

🎯 **Hikaru's mistake (preview of Step 4 diagnostic)**

If Hikaru writes: "*The correlation is 0.18, so the food doesn't cause aggression*", she has missed the critical value (0.18 vs n=100 critical) and used causal language. Always compare r_s to critical value and use **"due to chance"** not **"causes"**.

Step 5 — V / R / G evaluation

COURSES §6

Validity (was the test a true measure?)

Read §6 first. Fill in one weakness for your own study + link to WHY it matters.

Validity weakness #1:

WHY it matters (link to which kind of validity): (face / ecological / concurrent / third variable)

Validity weakness #2:

WHY it matters:

Reliability (would it give the same result again?)

Reliability weakness:

WHY it matters (link to standardisation / order effects / subjectivity):

Generalisability (can findings leave the lab?)

Generalisability weakness #1: (sample-related)

WHY it matters:

Generalisability weakness #2: (culture / sex / age related)

WHY it matters:

Compare grid — the 3 axes

Axis	Key question	Textbook critique of masculinity × aggression
V Validity	Were the variables a true measure?	Quiz is superficial; game measures gaming skill not real aggression; correlation cannot establish cause (third variable).
R Reliability	Were the procedures consistent?	Standardised but order effects, varied gap between tasks, varied prior gaming experience.
G Generalisability	Can we apply beyond the sample?	Small n=10; volunteer bias; Western industrialised; male-only → not women, not other cultures.

Step 6 — Improvements + 🔍 Hikaru #3

COURSES §7

Improvement checklist (textbook p.226)

Each weakness must be paired with an improvement. The improvement must address *that specific* weakness.

Validity improvement (link to which weakness on p.10)

WHY this fixes it:

Reliability improvement (link to which weakness)

WHY this fixes it:

Generalisability improvement (link to which weakness)

WHY this fixes it:

🔍 Diagnose Hikaru — improvement question (WPS02/01 Q2e)

Scenario — Hikaru's improvement

A student writes: ***"Hikaru should improve her study by using a bigger sample."***

Q1 · Is this a valid improvement? Why or why not?

Q2 · Suggest an improvement for the *closed question* "Were you aggressive today?" (be specific, link to validity)

Q3 · Suggest an improvement that would address the *single-school* generalisability weakness

📝 Self-check before moving on

An "improvement" that doesn't address a specific weakness scores 0. Always write: **weakness → improvement → WHY this fixes it**. The "why" is the second mark.

Step 7 — Report writing + 3 exam Qs

COURSES §8

Report skeleton

Fill the 3 sections from your own study. Use the textbook checklists on p.12 of the courseware.

Procedure section (8-point checklist (sample, apparatus, design, brief/debrief, location, individual/group, procedure + timings, controls))

Results section (raw data table, descriptive stats, scatter diagram + interpretation, Spearman's + significance)

Discussion section (conclusions, wider context, V/R/G, improvements, practical uses)

3 textbook exam questions (p.226) — write model answers

Q1 · Describe the procedure of your practical investigation (4 marks)

Mark scheme expects 4 distinct points: Sample · Apparatus & scoring · Design · Procedure.

Q2 · Explain whether the findings were statistically significant (2 marks)

Mark scheme: 1 mark for calculated vs critical; 1 mark for correct decision + hypothesis supported.

Q3 · Explain one validity improvement (2 marks)

Mark scheme: 1 mark for validity weakness; 1 mark for linked improvement.

🚀 **EXAM TIP (p.226)**
Treat your practical as the **named study**: AMSPRC + V/R/G = full marks.

Brain dump + answer key

FINAL CHECK

Brain dump — write everything you remember (5 min, no notes)

Before checking the answer key below, set a 5-minute timer and write everything you remember about Ch19 here.

Aim · hypothesis · operationalisation · sampling · ethics · controls · pilot · 6-step procedure · Spearman's rank · significance · V/R/G · improvements · 3 exam questions · AMSPRC

Answer key — verify your answers

p.2 — Spec 3.4 musts: (1) correlational study + descriptive stats + non-parametric test (Spearman's); (2) design decisions (co-variables, operationalise, ethics, hypothesis, controls); (3) scatter diagram; (4) Spearman's + statistical significance; (5) V/R/G; (6) report writing (procedure/results/discussion).

p.3 — Worked example (pp.216-217): Aim = "To investigate whether there is a link between the masculinity of a person and aggressive behaviour." H_1 = directional positive correlation between quiz score and game score. H_0 = "There will be no relationship... any relationship found will be due to chance." Co-variable 1 = masculinity quiz 0–100 (ordinal). Co-variable 2 = characters knocked down in video game 0–20 (ordinal).

p.4 — Hikaru Q1: "Were you aggressive today?" is **nominal** (yes/no). Spearman's needs at least ordinal. Q2: could be directional or non-directional; teacher rating supports either. Q3: 12 is below threshold; needs **no one under 16** + parental consent + child assent.

p.6 — Hikaru: Q1 sampling = single school → unrepresentative, cannot generalise. Q2 ethics = informed consent alone not enough; child assent required; teacher rating risks protection from harm. Q3 reliability = 12-year-olds interpret "aggressive" inconsistently.

p.8 — Spearman worked: $\Sigma d^2 = 112.5$; $n=10$; $r_s = 1 - (6 \times 112.5)/(10 \times 99) = 1 - 0.682 =$ **0.318**.

p.9 — Significance: calculated 0.318 < critical 0.564 → **not significant** → accept null. Model conclusion: "The calculated value of the Spearman's rank test was r = 0.318. This was less than the critical value of 0.564 for a one-tailed test at $p < 0.05$ with $n = 10$. Therefore the result is not significant and the null hypothesis can be supported, which states that there will be no relationship between masculinity and aggression, and any relationship found was due to chance."

p.10 — V/R/G textbook critiques: V — quiz superficial (face validity); game measures skill not aggression (ecological + concurrent); correlation ≠ cause (third variable). R — order effects; varied task gap; varied gaming experience. G — small n=10; volunteer bias; Western; male-only.

p.11 — Improvements: V — Rosenzweig Picture-Frustration Test for ecological; comprehensive masculinity scale. R — identical time slot. G — random/stratified sampling; cross-cultural; both sexes.

p.12 — Q1 procedure 4 marks: Sample + Apparatus + Design + Procedure (full in courseware §8). **Q2 significance 2 marks**: calculated vs critical (1) + correct decision + hypothesis supported (1). **Q3 improvement 2 marks**: validity weakness (1) + linked improvement (1).

📌 Final self-check before the exam

- I can write AMSPRC for my own practical from memory.
- I can write the model conclusion with all 7 ingredients.
- I can spot the closed-question / single-school / yes-no issues in Hikaru's design.
- I can give one V, one R, one G weakness and a linked improvement for each.
- I can describe my procedure in 4 distinct points (sample / apparatus / design / procedure).